

Relax Data Science Take-Home Challenge

Overview

This analysis identifies factors associated with user adoption using engagement and user attribute data. An **adopted user** is defined as a user who logged into the product on **3 separate days within any 7-day period**. Two datasets were provided: (1) **takehome_users.csv** (user attributes) and (2) **takehome_user_engagement.csv** (login activity).

Approach

- Define Adoption:** For each user, I checked for ≥ 3 unique login days within any rolling 7-day window and created the binary target variable `adopted_user` (1 = adopted, 0 = not adopted). Adopted users comprised 13.8% of the cohort, revealing an imbalanced dataset.
- Feature Engineering:** The following features were used in the model: (1) **creation_source** (how user signed up), (2) **was_invited** (binary variable derived from `invited_by_user_id`), (3) **opted_in_to_mailing_list**, (4) **enabled_for_marketing_drip**, (5) **org_size** (number of users in organization—derived from `org_id`)
- Modeling: Logistic Regression** was used as the classification model. It provides a strong baseline and, since it is interpretable, can be used to identify which factors predict future user adoption. The model included **one-hot encoding** for categorical variables, standardization for numeric features, and class balancing to address the class imbalance between adopted and non-adopted users. Binary features were passed through without transformation.

Results

Model Performance: The model shows weak predictive power (ROC-AUC ≈ 0.60), indicating limited ability to distinguish between adopted and non-adopted users.

It achieved high recall for adopted users (0.76), meaning it successfully identified most adopters, but with low precision (0.17) indicating many false positives. This tradeoff reflects class imbalance and the use of class weighting.

Overall, the results suggest that the available features provide limited predictive signal, and that early user behavior or engagement patterns may be more informative than user attributes available at signup.

Key Predictors of Adoption: The analysis indicates that how users join the platform is a key predictor of adoption. Users who were invited or signed via Google authentication were more likely to adopt, while those joining through personal projects were significantly less likely to engage. Larger organization size was associated with lower adoption, suggesting reduced individual engagement in larger groups. Marketing-related features showed minimal impact.

FEATURE	EFFECT ON ADOPTION	APPROX. CHANGE IN ODDS
Google signup	↑ Higher likelihood	+32%
Invited users	↑ Higher likelihood	+30%
Personal projects	↓ Lower likelihood	-45%
Larger org size	↓ Lower likelihood	-27%

Limitations

- Limited feature set: only user attributes available at signup were used; early engagement signals (e.g., first-week activity) were not included
- No hyperparameter tuning was performed, as the focus was on establishing a clear and interpretable baseline model
- Time-based features derived from post-signup activity were excluded to avoid data leakage

Next Steps

With more time, I would:

- Incorporate early engagement signals (e.g., first-week activity), which are likely stronger predictors of adoption
- Evaluate more flexible models (e.g., Random Forest, Gradient Boosting) to capture nonlinear relationships
- Explore interaction effects between signup method and organizational context
- Conduct cohort analysis by signup source or organization to better understand behavioral differences